

1. What is Express.js?

Express.js is a framework of **Node.js**.

It helps us create:

- Web Server
- APIs
- Backend Applications

Why do we use Express?

- Easy to learn
- Less code
- Fast development
- Easy routing
- Can send JSON data

Example

```
const express = require("express");
```

```
const app = express();
```

Explanation:

- `require("express")` → Imports Express.
 - `express()` → Creates an Express application.
 - `app` → Our server.
-

2. What is Routing?

A **Route** tells the server what to do when a user visits a URL.

Syntax

```
app.get(path, callback);
```

Example

```
app.get("/", (req, res) => {  
  res.send("Welcome");  
});
```

If the user opens

`http://localhost:3000/`

Output

Welcome

Example Routes

```
app.get("/", (req, res) => {  
  res.send("Home Page");  
});
```

```
app.get("/about", (req, res) => {  
  res.send("About Page");  
});
```

```
app.get("/contact", (req, res) => {  
  res.send("Contact Page");  
});
```

3. HTTP Methods

HTTP methods tell the server **what action** to perform.

The most common methods are:

- GET
 - POST
-

4. GET Method

GET is used to **read or fetch data**.

Example

```
app.get("/students", (req, res) => {  
  res.send("Student List");  
});
```

URL

localhost:3000/students

Output

Student List

Uses of GET

- Get all students
 - Get products
 - Get user details
 - Search data
-

5. POST Method

POST is used to **send new data** to the server.

Example

```
app.post("/students", (req, res) => {  
  res.send("Student Added");  
});
```

Example

POST /students

Output

Student Added

Uses of POST

- Add Student
 - Register User
 - Login
 - Add Product
-

GET vs POST

GET	POST
Read data	Send new data
Data in URL	Data in Request Body
Safe	Used to create data
Faster	Slightly slower

6. JSON Response

JSON stands for

JavaScript Object Notation

JSON is used to send data between the server and client.

Example

```
const student = {  
  id: 1,  
  name: "Mukund",  
  course: "CSE"  
};  
  
res.json(student);
```

Output

```
{
  "id":1,
  "name":"Mukund",
  "course":"CSE"
}
```

Why JSON?

- Easy to read
 - Easy to send
 - Used by almost every API
-

7. Request and Response

Request (req)

The client sends a request to the server.

Example

GET /students

Response (res)

The server sends data back.

Example

```
res.send("Hello");
```

or

```
res.json(data);
```

8. Route Parameters

Used to get data from the URL.

Example

```
app.get("/students/:id", (req, res) => {

  const id = req.params.id;

  res.send(id);

});
```

URL

localhost:3000/students/5

Output

9. Query Parameters

Used for searching or filtering.

Example

```
app.get("/search", (req, res) => {  
  
  const name = req.query.name;  
  
  res.send(name);  
  
});
```

URL

localhost:3000/search?name=Mukund

Output

Mukund

10. Nodemon

Nodemon automatically restarts the server whenever we save the file.

Without Nodemon

Change Code

↓

Stop Server

↓

Run Again

With Nodemon

Change Code

↓

Save File

↓

Server Restarts Automatically

Install

```
npm install -g nodemon
```

Run

```
nodemon app.js
```

11. Student API

Example

```
const students = [  
  { id: 1, name: "Mukund" },  
  { id: 2, name: "Rahul" }  
];
```

Routes

GET /students

Returns all students.

GET /students/1

Returns one student.

POST /students

Adds a new student.

12. Product API

Example

```
const products = [  
  { id: 1, name: "Laptop" },  
  { id: 2, name: "Phone" }  
];
```

Routes

GET /products

Returns all products.

GET /products/2

Returns one product.

POST /products

Adds a new product.

13. Important Functions

```
express()
```

Creates an Express application.

`app.get()`

Handles GET requests.

`app.post()`

Handles POST requests.

`res.send()`

Sends text or HTML.

`res.json()`

Sends JSON data.

`app.listen(3000)`

Starts the server on port **3000**.

14. Interview Questions

Q1. What is Express.js?

Answer: Express.js is a Node.js framework used to build web servers and APIs quickly.

Q2. What is Routing?

Answer: Routing maps a URL to a specific function in the server.

Q3. What is the difference between GET and POST?

Answer:

- GET is used to read data.
 - POST is used to send or create new data.
-

Q4. What is JSON?

Answer: JSON (JavaScript Object Notation) is a lightweight format used to exchange data between client and server.

Q5. Why do we use Nodemon?

Answer: Nodemon automatically restarts the server whenever the code is saved, making development faster.